

Safety Gear Monitoring System

Introduction

The Safety Gear Monitoring System is an intelligent computer vision solution designed to detect whether personnel are wearing the required safety gear such as helmets and vests using object detection models.

Objective

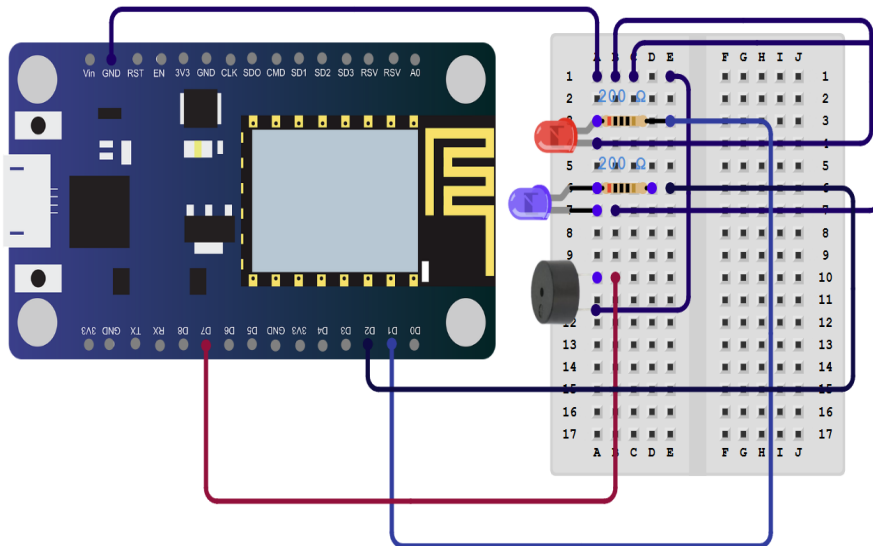
To develop a real-time system that detects and monitors the usage of safety gear (helmet and vest) using YOLOv8 and provides alerts and reports based on the compliance status.

Methodology

1. **Data Collection**: Gather images with variations of people wearing and not wearing helmets and vests. 2. **Model Training**: Use YOLOv8 to train a custom model with four classes – helmet, no_helmet, vest, no_vest. 3. **Real-Time Detection**: Employ PyQt5 for GUI and DeepSORT for tracking objects and updating class-wise counts. 4. **Violation Logging**: Track safety violations and save logs/videos for review.

Hardware Setup

The system utilizes a basic camera (e.g., webcam/CCTV), a standard PC (GPU for better performance), and optionally an alert system. Below is the hardware setup image:



Results and Conclusion

The system achieves accurate real-time tracking with consistent ID assignment. It logs safety violations, updates per class count (e.g., number of workers with or without helmet/vest), and creates hourly backups of video feeds.

Conclusion

This project effectively demonstrates the application of computer vision in industrial safety. It can be extended further for attendance monitoring or zone-based safety compliance.

Reference

- YOLOv8 by Ultralytics - DeepSORT for object tracking - PyQt5 for GUI - OpenCV for video processing